CURRICULUM VITAE



NAME: ANUBHA DEY

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Research Gate: Anubha Dey
Web of Science: Anubha Dey - Web of Science Researcher Profile

EDUCATION

- Postdoctoral Fellow at Koita Centre for Digital Health, IIT Bombay, 2026
- Ph.D., University of Hyderabad, 2025, 9.14/10
- Master of Technology (M.Tech.), Anna University, Chennai, 2017-2019, 9.38/10
- Bachelor of Technology (B.Tech.), Maulana Abul Kalam Azad University of Technology, West Bengal, 2012-2016, 8.41/10

Academic Goals

My long-term goal is to establish myself as an independent researcher in academia, where I can pursue my passion for advancing human health and disease research. I am eager to share my knowledge and experience with future generations by teaching and mentoring young minds, inspiring them to become the next leaders in scientific discovery and innovation.

Publications:

- Dey, Anubha Kiran, Manjari. "Machine learning approaches for the identification of genetic interactions" in the Artificial Intelligence (AI) in Cell and Genetic Engineering Book. Publisher: Springer Nature

  My contribution: I reviewed different machine learning models to predict genetic interactions and discussed their advantages and limitations

- Dey, Anubha Kiran, Manjari. "R-based protocols to predict synthetic lethal interactions in cancers using machine-learning tools" in Artificial Intelligence (AI) in Cell and Genetic Engineering Book". Publisher: Springer Nature

  My contribution: I executed two machine learning models and exhibited each in R Markdowns.

- Dey, Anubha., Mudunuri Suresh., Kiran, Manjari., (2024), MAGICAL: A multi-classifier to predict synthetic lethal and viable interactions using a protein -protein interaction network, PLOS Computational Biology (MAGICAL: A multi-class classifier to predict synthetic lethal and viable interactions using protein-protein interaction network - PubMed)

  My contribution: PhD work in which I developed a multi-class machine learning model to predict genetic interactions. Utilized network properties as features for model training. Also carried out survival analysis to depict patient response to genetic interactions

- Basu, Koushiki., Dey, Anubha., Kiran, Manjari., (2024), CaTCH: Calculating transcript complexity of human genes, Methodx (CaTCH: Calculating transcript complexity of human genes - PubMed)

  My contribution: Developed a binary class random forest machine learning model for predicting transcript complexity

- Bond Marielle L., Davis Eric S., Quiroga Ivana Y., Dey Anubha., Kiran Manjari., Love Michael I., Won Hyejung., Phanstiel Douglas H. (2023), Chromatin loop dynamics during cellular differentiation are associated with changes to both anchor and internal regulatory features, Genome Research (Chromatin loop dynamics during cellular differentiation are associated with changes to both anchor and internal regulatory features - PubMed)

  My contribution: Developed a multi-class random forest machine learning model to predict differential looping utilizing features such as anchor CTCF, interior ATAC, and interior H3K27ac.

- Basu, Koushiki., Dey, Anubha., Kiran, Manjari., (2023), Inefficient splicing of long non -coding RNAs is associated with higher transcript complexity in human and mouse, RNA Biology, Taylor & Francis,20,1,563-572 (Inefficient splicing of long non-coding RNAs is associated with higher transcript complexity in human and mouse - PubMed)

  My Contribution: I was involved in technically supporting the work in data analysis

- Kondapally, Mavika., Dey, Anubha., Golagana, Harshita., Kiran, Shashi., Kiran, Manjari., (2024), Deubiquitinases as prognostic biomarkers and potential drug targets for gynecological cancers, International Journal of Clinical Oncology (Deubiquitinases as prognostic biomarker and potential drug target for gynecological cancers | International Journal of Clinical Oncology)

  My contribution: I was involved in technically supporting the work in data analysis

- Jindal, Haneesh., Dey, Anubha., Teja, Pranav., Mudunuri, Suresh., Kiran, Manjari., (2024), CHOLAR: Characterization of LncRNA from raw reads, F1000 (The 22nd International Conference on Bioinformatics (InCoB... | Documents | F1000Research)

  My contribution: I was involved in technically supporting the work in data analysis

Manuscript under preparation

- MAGICAL-DB: A resource for negative and positive interactions in cancer

  Dey, Anubha., Mudunuri Suresh., Kiran, Manjari., MAGICALdb: A resource for negative and positive interactions in cancer

  My contribution: Identified and deposited all the predicted SL and SV interactions in the MAGICAL database.

- Identification of tumorigenic cell types and synthetic lethal pairs for effective targeted therapy in glioblastomas

  Ghalgond, Shrithika., Dey, Anubha., Kiran, Manjari., Identification of tumorigenic cell types and synthetic lethal pairs for effective targeted therapy in glioblastomas

  My contribution: Identified clusters of different cell types in single-cell RNA seq data. I have further utilized cSL load to identify the tumorigenic cell types.

- Pan-cancer identification of sex-biased prognostic genes and markers

  Satpathy, Sarthak., Basu, Koushiki., Dey, Anubha., Kiran, Manjari., Pan-cancer identification of sex-biased prognostic genes and markers

  My contribution: I was involved in technically supporting the work in data analysis.

Technical Skills

R language

Shell scripting

Python basics

AWARDS/ HONORS

- Selected for **Young Scientist Awards**, in the category, [IBS Award: Best oral presentation in Medical Biotechnology / Software in Medicine], at the International Conference on Medical Biotechnology ICoMB at Amity Noida on 7-9th January 2026
- Invited for a talk at International Conference on Medical Biotechnology (ICoMB), at Amity Noida on 7-9th January 2026
- Selected for a talk at HySci at IIT Hyderabad on 20th April 2024. HySci is a scientific conference where the brightest minds from diverse backgrounds share visions and discoveries
- Best Oral Presentation Award at BioAnveshna: Frontiers in Basic Biology, Biotechnology & bioinformatics", on 15-18th February 2024, at the School of Life Sciences, University of Hyderabad. Felicitated by Nobel Laureate, Sir Richard J. Roberts
- Selected and invited to deliver a lightning talk at the 4th Indian Cancer Genome Atlas (ICGA) conference, IISER Pune 2023
- Best Poster Presentation Award at the 16th International Cancer Symposium on "Translational Chemoprevention and Brain Storming" under the theme, Artificial Intelligence and Machine Learning in Cancer Research, held on 18-19th November 2022 at Jawaharlal Nehru University
- Qualified for the Joint Entrance Screening Test (JEST) 2019. It is a national entrance test in theoretical computer science
- Served as the School President and Best Leader of the student council

WORKSHOPS/ CONFERENCES:

- Presented virtual poster "MAGICAL: A Multi-Class Classifier to Predict Synthetic Lethal and Viable Interactions Using Protein-Protein Interaction Network" at the International Conference on Research in Life-Sciences & Healthcare (ICRLH) held in Seoul, Korea (South) on 23 - 24 May, 2024
- Participated in the workshop on "Statistical Machine Learning for Biologists" held at the Centre for Molecular Simulation and Design (CMSD) from 19-22 December 2022, organized by DBT-Centre for Microbial Informatics (DBT-CMI), University of Hyderabad
- Presented virtual poster iMCLAPE: A multi-class classifier on epistatic interaction, on Biological Data Science, held on 09-12th November 2022, at Cold Spring Harbour Laboratory (CSHL), New York, United States
- Presented poster iMCLAPE: A multi-class classifier on epistatic interaction on the 12th India-Japan Science and Technology Conclave, International Conference on Frontier Areas of Science and Technology (ICFAST-2022), at the University of Hyderabad
- Successfully completed the online workshop on "Applications of machine learning techniques in biology using WEKA" on 03-04th September 2022 by Prof. M Michael Gromiha, IIT Madras
- Presented partial work of the final semester project at the 32nd annual scientific conference at Vivekananda Institute of Medical Sciences, 23rd-25th November 2018
- Attended ROBO AVR, a workshop on Automated Vehicle Robotics recognized by Science Park, Korea in association with MSRDS-UG & AIESEC IIT Delhi, held at NIT Durgapur

Work Experience: Served Tata Consultancy Services as an "Assistant System Engineer- Trainee" for 11 months from 03/10/2016 to 07/09/2017

Membership

- Member of the International Society for Computational Biology (ISCB)
- Member of the Society for Biotechnologists India (SBTI)